Computer Vision | Grading rubric

CS418 · Midterm exam, Version A · Fall 2026 · Instructor copy

5 parts / 100 marks: P1 MCQs 40 · P2 Q21–22 20 · P3 Q23–24 20 · P4 Q25 10 · P5 Q26 10

MCQs Q1–20: 2 marks per correct single answer; 0 otherwise. No negative marking.

Q1–5	A B C B B
Q6–10	D A C C A
Q11–15	C B A D D
Q16–20	A B D D C

Q21 · 10 Camera output	**(a) 3:** Shape (30, 50, 3) [1]; 4,500 bytes [1]; swapped dimensions preserve size but change row/column meaning and bounds [1]. **(b) 3:** MAD to A = 5, to B = 1 [1]; B wins although Q depicts A [1]; brightness misleads raw-pixel matching [1]. **(c) 4:** Vary lamp brightness [1]; same intact parcel and pose [1]; fixed camera/settings [1]; explain that pixel changes can reflect lighting without damage [1].
Q22 · 10 Image operations	**(a) 3:** Histogram [4,4,4,4] [1]; mapping 0→0, 1→2, 2→3, 3→3 [1]; new histogram [4,0,4,8] [1]. **(b) 3:** Exact recovery not always possible [1]; 2 and 3 both map to 3 [1]; original level at clipped pixels is lost [1]. **(c) 4:** Reversal orders agree [1]: pointwise T commutes with rearrangement [1]. A then T = 3; T then A = 2.5 [1]; explain clipping makes averaging order matter [1].
Q23 · 10 Gradients / damage	**(a) 3:** Gx = 18 [1]; Gy = 0 [1]; magnitude = 18 [1], with signed working. **(b) 4:** Add 5: (18,0) [1], since kernels sum to zero [1]. Multiply by 2: (36,0) [1], by linearity [1]. **(c) 3:** Rip conclusion unsupported [1]; printing also creates gradients [1]; propose a specific comparison, e.g. aligned intact label under comparable lighting [1].
Q24 · 10 Corners / matching	**(a) 4:** R(A) = 30.24 [1]; R(B) = −36 [1]; choose A [1]; B varies strongly in only one direction [1]. **(b) 3:** New response 16R [1]; determinant and squared trace both scale by 16 [1]; sign unchanged [1]. **(c) 3:** P: 0.9, reject; Q: 0.5, accept [1]; ratio test does not prove identity [1]; geometric consistency check [1].
Q25 · 10 Linear classifier	**(a) 4:** LEFT = 1 [1]; RIGHT = 2 [1]; predict RIGHT [1]; scores are not probabilities (e.g. 2 > 1 / sum ≠ 1) [1]. **(b) 3:** Softmax probabilities unchanged [1]; prediction unchanged [1]; common exp(7) factor cancels [1]. **(c) 3:** RIGHT remains predicted and wrong [1]; true-class probability rises, so cross-entropy falls [1]; loss improves despite unchanged correctness [1].
Q26 · 10 Model / learning rate	**(a) 4:** Choose epoch 6 [1]; minimum validation loss 0.42 [1]; later divergence indicates overfitting [1]; untouched test set used after selection [1]. **(b) 2:** New weight −0.25 [1]; gradient descent subtracts the positive gradient term [1]. **(c) 4:** A: weight −1; B: −3 [1]; losses 0.5 and 4.5 versus initial 2 [1]; sign change alone is inconclusive [1]; A improves, B worsens [1].

Partial credit: [1] denotes one mark. Accept equivalent valid methods and explanations. Award correct steps; do not penalize the same arithmetic error again in later consistent work. For paired results under one mark, award 0.5 for one correct result (Q21b, Q22c, Q24c, Q26c). Accept sensible rounding and clear reasoning despite minor language errors.

Basis: CS418 Midterm Exam, Version A (22 October 2026 revision; saved 6 November 2026), and matching instructor grading guide.